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Machine Learning for Privacy Threat Classification: A Systematic Review

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Abstract

Privacy threat classification plays a vital role in protecting sensitive information in today's digital landscape, especially across domains such as the Internet of Things (IoT), smart devices, and cloud-based services. As traditional rule-based security mechanisms delay processing the limitations of under increasing data volume and threats, machine learning (ML) techniques have emerged as promising solutions. Despite significant advancements, these ML driven systems face challenges including data heterogeneity, sparsity, unlabeled data, class imbalance, evasion attacks, and concept drift which limit their performance in real-world applications. Existing surveys on this topic have largely focused on narrow domains such as IoT or federated learning, lacking limitations and omitting critical aspects such as how algorithms works, deployment scalability, and cross-domain applicability. In response, this systematic literature review aims to identify the available machine learning algorithms for privacy treat classification, limitations of the algorithms and evaluate current mitigation strategies. By addressing three core research questions, the study provides existing work,

highlights unresolved challenges of ML algorithms, and proposes future research directions to mitigate these limitations

Keywords: Machine Learning, Privacy Threats, Classification

1 Introduction

The classification of privacy threats is a critical component in protecting sensitive data, particularly within the expanding digital system of the Internet of Things (IoT) , smart devices, and cloud-based services[1–4]. As data becomes more distributed and realtime, traditional rule-based security mechanisms struggle to cope with the growing complexity and dynamic nature of modern threats [3, 5]. Machine learning (ML) techniques have been increasingly employed to automate the classification of privacy threats by learning from vast datasets and identifying patterns indicative of malicious behavior [6].

The classification of privacy threats involves categorizing various types of threats that compromise the confidentiality, integrity, or availability of data. These may include unauthorized access, data leakage, inference attacks, and adversarial manipulations such as poisoning or evasion attacks [7, 8]. ML models help distinguish between benign and malicious data patterns, enabling real-time detection and proactive response to emerging privacy risks. For example, supervised learning algorithms such as Support Vector Machines (SVMs), Decision Trees, and Random Forests have been applied to detect specific types of privacy threats by classifying labeled instances of known attacks [9]. Meanwhile, unsupervised models such as clustering and anomaly detection techniques are used to uncover previously unknown threats by identifying deviations from normal system behavior[10–14].

Despite these advances, privacy threat classification remains challenging due to the highly imbalanced and heterogeneous nature of privacy-related datasets, as well as the presence of adversaries capable of adapting their tactics to avoid detection [15–17]. Furthermore, issues such as concept drift, where the statistical properties of threats evolve over time, require continuous retraining and adaptation of ML models [18]. These limitations underscore the need for robust and explainable ML frameworks that can not only classify threats accurately but also provide transparency and resilience in real-world deployments [16, 19]. As privacy threats continue to evolve in complexity and sophistication, the role of ML-based classification systems will become even more crucial. However, a comprehensive understanding of the limitations and vulnerabilities of these systems is essential to develop secure, adaptive, and reliable solutions in the era of pervasive computing.

Several surveys have been conducted to explore the role of machine learning (ML) in privacy threat classification, each providing valuable insights while also exhibiting notable limitations. For example, Al-Garadi et al. [6] offers a comprehensive review of ML and deep learning methods for privacy protection in IoT environments, but their focus is limited to IoT-specific scenarios, lacking a broader cross-domain perspective.

Similarly, Mothukuri et al. [9] investigate privacy challenges in federated learning systems, highlighting vulnerabilities in decentralized settings, but do not extend their findings to traditional centralized ML models. Shafee et al. [20] examine realistic adversarial threats in network intrusion detection systems, providing practical guidance, although their study remains confined to intrusion detection without addressing other forms of privacy threats such as data leakage or inference attacks. Furthermore,Yuan et al. [21] discuss adversarial examples and mitigation techniques in deep learning, but their work is not directly tailored to privacy threat classification. A common limitation across these surveys is the absence of limitations of ML algorithms in terms of data, algorithmic design, and deployment context. Furthermore, many of these reviews lack empirical validation and fail to address the scalability and transparency challenges faced in real-world applications. These gaps underscore the need for a comprehensive, cross-domain survey that systematically classifies the limitations of ML algorithms in privacy threat classification and evaluates mitigation strategies. Understanding these limitations is crucial to advancing the state of ML for privacy protection. This systematic literature review aims to find limitations of the exsisting algorithms and future directions.

The remainder of the paper is structured as follows: Section 2 Objectives of the survey. Section 3 metholodology of the survey. Section 4 presents an overview of the reviewed literature. Section 5 offers a results. Section 6 discusses mitigation strategies and outlines open challenges and future directions. Section 7 concludes with a summary of findings and implications.

2 Objectives

In this systematic review, we aim to identify available machine learning algorithms for privacy threat classfications, the limitations of machine learning (ML) algorithms used in privacy threat classification, particularly within the Internet of Things (IoT) environments and the how to overcome these limitations. The growing complexity of IoT systems and the vast amounts of data they generate present unique challenges for ML-based privacy detection systems, such as imbalanced datasets, data heterogeneity, and adversarial manipulation. By analyzing these limitations, we assess their impact on the efficacy and reliability of ML models in real-world applications. This analysis will highlight the factors that hinder the accurate identification of privacy threats, such as unauthorized access, data leakage, and inference attacks. Furthermore, the review will provide an overview of mitigation strategies, such as data preprocessing techniques, adversarial training, and model robustness enhancements, which aim to address these challenges. Lastly, we propose future research directions to enhance the robustness of ML algorithms in privacy threat classification, focusing on improving scalability, transparency, and adaptability to evolving threats in dynamic IoT environments. Following are the identified objectives.

- Identify the available ML algorithms in privacy threat classification within IoT environments.
- Identify the limitations of ML algorithms in privacy threat classification within IoT environments.

- Provide insights into existing mitigation strategies and propose research directions for enhancing the robustness of these models.

Reference	Year	Description	Limitations
Al-Garadi et al. [6]	2023	A survey of ML and deep learning methods for privacy protection in IoT environments.	Focuses only on IoT; lacks broader cross-domain perspective.
Mothukuri et al. [9]	2021	Investigates privacy challenges in federated learning systems, highlighting vulnerabilities in decentralized settings.	Does not cover centralized ML models; narrow focus on federated learning.
Shafee et al. [22]	2023	A comprehensive survey of realistic adversarial threats in network intrusion detection systems.	Confined to network intrusion; does not address other privacy threats like data leakage or inference attacks.
Yuan et al. [21]	2019	Discusses adversarial examples and mitigation techniques in deep learning, with a focus on security concerns.	Not specifically targeted at privacy threat classification.
Lockwood et al. [23]	2021	Provides guidelines, tools, and checklists for authors conducting systematic reviews.	Not focused on ML or privacy threat classification; general review methodology.
Pei et al. [24]	2022	Focuses on federated learning and privacy, emphasizing machine learning algorithms in privacy contexts.	Limited to federated learning; does not address traditional ML frameworks.
Manoharan et al. [15]	2024	A survey of insider threat detection using ML algorithms, specifically in imbalanced datasets.	Primarily focused on insider threats; does not consider all privacy-related threats.
Jagielski et al. [16]	2020	A survey of subpopulation data poisoning attacks in machine learning and their impact on privacy.	Focused on poisoning attacks; does not consider all privacy attack vectors.

Table 1 Survey papers on ML-based privacy threat classification and their limitations.

2.1 Research Questions

The research questions for this systematic review are designed to address the key limitations identified in existing survey papers on machine learning (ML) for privacy threat classification. Firstly, we aim to identify the exsisting ML algorithms in this domain, particularly those highlighted across surveys, which often focus on specific contexts such as IoT or federated learning, thereby missing a broader, cross-domain perspective. Secondly, we seek to explore the limitations of these algorithms, considering how algorithms are work that existing surveys fail to fully capture. This includes the problem of concept drift, where models need constant retraining to adapt to new types of attacks. Lastly, we will investigate the mitigation strategies proposed in the literature, critically examining their effectiveness in addressing these challenges. Existing surveys have been limited in their focus, often lacking empirical validation and practical insights into real-world applicability, which we aim to remedy by proposing research directions that can further enhance the robustness and generalization of ML models for privacy threat classification in dynamic environments.

1. What are the existing ML algorithms in privacy threat classification?
2. What are the limitations of ML algorithms in privacy threat classification?
3. What strategies have been proposed to overcome these limitations?

3 Methodology

This study follows the process proposed by Lockwood et al. [23] in their work, to conduct a systematic literature review. The process consists of several key steps to ensure a structured and comprehensive approach to reviewing the literature.

1. **Objective and Research Questions:** The first step involves identifying a clearly structured research problem and setting the research objectives. This step ensures that the review addresses a specific, focused question that guides the entire review process.
2. **Eligibility of Studies:** In this phase, we determine the criteria for selecting the studies to be included in the review. This includes defining the scope, topic relevance, and time frame of the studies, ensuring that only the most pertinent research is considered.
3. **Primary Study Selection:** A detailed protocol is then developed to guide the extraction of relevant studies. This protocol specifies the search strategy, databases to be used, and keywords, ensuring that the selection process is transparent and reproducible.
4. **Quality Assessment:** To ensure that only high-quality literature is included, clear inclusion and exclusion criteria are set. This step involves evaluating each selected study against predefined quality benchmarks to assess its reliability and relevance to the research questions.
5. **Data Extraction:** Relevant data from the selected studies are then extracted. This includes gathering information such as study design, methodologies, key findings, and other factors that contribute to answering the research questions.

6. **Data Analysis:** The extracted data are then analyzed. The analysis involves synthesizing the findings from different studies, presenting them in both narrative and tabular forms to offer a comprehensive overview of the state of the field. This step aims to identify patterns, trends, and gaps in the current body of knowledge.

3.1 Eligibility of studies

This systematic review aims to identify the the existing machine leaning algorithms and limitations of ML algorithms in privacy threat classification and evaluate the strategies proposed to overcome these limitations. To ensure the inclusion of relevant studies, a structured and thorough search was performed across several academic databases, including IEEE Xplore, ACM Digital Library, SpringerLink, Google Scholar, Web of Science, and Scopus. The eligibility criteria were designed to capture studies that provide insights into the limitations of ML models in IoT privacy applications, as well as the effectiveness of mitigation strategies. Studies published between 2015 and 2025 were considered, focusing on experimental validation, real-world datasets, and context-specific analysis of privacy threat detection in IoT environments. The search was conducted using advanced search filters, such as year-wise and citation-wise filtering, to ensure comprehensive coverage of relevant papers as in Table 3.1.

The selected studies were required to find existing ML algorithms and explicitly discuss the limitations of ML-based privacy detection systems. Furthermore, only studies that proposed mitigation strategies to overcome these limitations were included, with particular attention paid to their effectiveness in real-world applications.

Table 3.1 shows the summary of the databases used and the number of papers retrieved through various search filters.

Database Name	**Advanced Search Provided**	**Year-wise Filtering Provided**	**Citation-wise Filtering Provided**
IEEE Xplore	Yes	Yes	Yes
ACM Digital Library	Yes	Yes	Yes
SpringerLink	Yes	Yes	Yes
Google Scholar	Yes	Yes	Yes
Web of Science	Yes	Yes	Yes
Scopus	Yes	Yes	Yes

Table 2 Search and Filtering Features of Each Database

3.2 Primary study selection

The overview of primary study selection is described in the following section and it contains 3 stages. Stage 1 is for the database selection, stage 2 is for strategic keyword search, and stage 3 is for filtering.

Database Name	Number of Papers
IEEE Xplore	120
ACM Digital Library	85
SpringerLink	75
Google Scholar	300
Web of Science	95
Scopus	110

Table 3 Number of Papers Retrieved from Each Database

Stage 1:Database selection; To obtain comprehensive research articles relevant to the systematic review, we selected the databases listed in the 3.1 considering the established standards and filtering mechanisms provided by such databases. The selected databases according to the filtering mechanisms provided are listed in Table 3.1.

Stage 2:Once keywords and their synonyms were identified, the search string was defined and used to search reputable online databases. This approach aims to include the most relevant research papers, ensuring the quality of the study's results. After the selection of databases, an advanced search was performed using the search string determined. The search was applied within the title, abstract, and keywords of the article.

```
"privacy threat classification" OR "privacy threat detection" OR "privacy
protection" OR "privacy risks" OR "privacy concerns"
AND
"Machine Learning" OR "ML" OR "Deep Learning" OR "Neural Networks" OR
"Supervised Learning" OR "Unsupervised Learning" OR "Reinforcement Learning"
AND
"Limitations" OR "Challenges" OR "Issues" OR "Drawbacks" OR "Limitations of
ML" OR "ML Constraints"
AND
"Mitigation" OR "Mitigation Strategies" OR "Overcoming limitations" OR
"Improving performance" OR "Robustness of ML models"
```

The search was conducted in databases including IEEE Xplore, ACM Digital Library, SpringerLink, Web of Science, Scopus, and Google Scholar. The advanced search was applied using the defined search string across the title, abstract, and keywords of the articles to capture the most relevant studies for the systematic review.

Stage 3: Filtering; Among the large numbers of papers selected from the advanced query search, the following filtering approaches were applied to extract the relevant and most recent articles in the area of study. Research papers were selected within the years between 2015 and 2025 which are restricted to the subject area of Computer Science. After going through stage 1, 2 and 3, results in Table 3 were obtained.

3.3 Quality assessment for inclusion/exclusion

This stage defines the exclusion and inclusion criteria for the systematic review of the article. This stage follows 3 criteria to decide the inclusion and exclusion of the papers in the systematic review based on the full text contents in the papers and the nature of publication. The summary of the criteria considered is shown in 3.3. Criterion 1: In the review, papers published in journals and conferences between the years 2015 and 2025 were included and others were excluded from the review. Criterion 2: Included papers written in English and excluded the others. Criterion 3: Included the full text articles from conferences and journals searched according to the query. Duplicated papers from each database were eliminated, and from the articles, unpublished and working papers, dissertations, short papers were excluded. Criterion 4: Papers lacking a clear indication of the research question in their title or abstract. Papers not relevant to classification are eliminated from the review.

Criteria	Value
1.	Included papers published in years between 2015–2025 Excluded papers published before 2015
2.	Included papers written in English Excluded non-English articles
3.	Included Journals and Conferences full text articles in the selected databases Excluded duplicate reports of the same study listed under different databases Excluded unpublished working papers, dissertations, short papers
4.	Included papers relevant to any of RQ1, RQ2, or RQ3 Excluded papers not relevant to privacy treats and have not addressed limitations Excluded studies that are not related to the research questions

Table 4 Inclusion and Exclusion Criteria

Q	Quality Assessment Question
Q1	Is the aim of the study clearly stated?
Q2	Are the methods clearly described and reflect the problem addressed? The privacy treat classification alleviation approach is clearly mentioned as a direct method or as a part of the overall solution.
Q3	Have they reviewed related studies extensively to support their solution and highlight its novelty?
Q4	Have they evaluated their implementation with appropriate datasets using state-of-the-art methods and metrics?

Table 5 Quality Assessment Criteria

Following the initial primary study selection process, a total of 785 articles were retrieved from five databases. Applying the inclusion and exclusion criteria based on criterion 1 and 2 reduced this number to 658. Subsequent refinement using criterion 3, which excluded duplicate entries, dissertations, short papers, and working papers, resulted in a narrowed pool of 584 papers.

Next, criterion 4 was applied, which involved filtering papers by examining their titles and abstracts to identify those relevant to recommender systems and the sparsity issue. This step reduced the set to 485 papers.

To ensure the relevance and focus of the remaining papers, an additional screening phase was conducted by closely examining whether the research questions were explicitly addressed in the title or abstract. Papers that clearly articulated their research questions in these sections were retained for review, resulting in a final set of 102 high-quality articles. All other papers were excluded.

These statistics are presented in Table 6. Figure 1 illustrates the distribution of selected articles by publication year following the multi-stage filtering process.

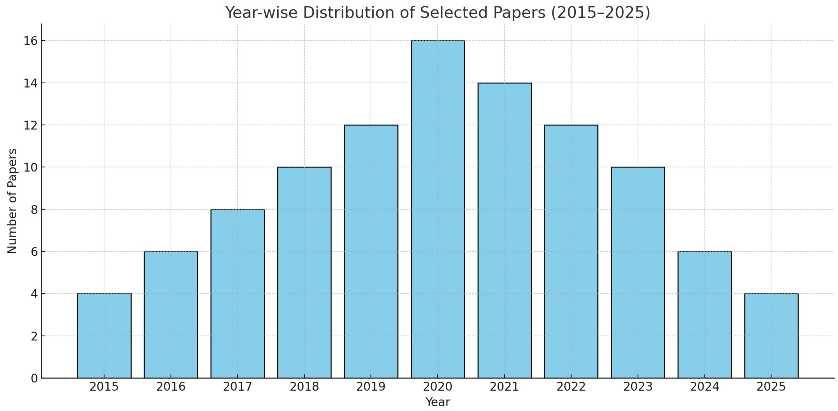

Fig. 1 Year-wise Distribution of Selected Papers (2015–2025)

3.4 Data extraction

For a systematic literature review on the limitations of machine learning (ML) algorithms in privacy threat classification, the data extraction process focuses on gathering relevant information from existing studies to address the following three research questions:

A comprehensive analysis was carried out on the selected studies to investigate the current landscape of machine learning (ML) algorithms applied to privacy threat classification. The primary aim was to address Research Question 1 (What are the existing ML algorithms in privacy threat classification?), Research Question 2 (What are the limitations of ML algorithms in privacy threat classification?), and Research Question 3 (What strategies have been proposed to overcome these limitations). Data were systematically extracted and synthesized to identify the commonly used ML algorithms, including supervised, unsupervised, and deep learning approaches. The limitations of these algorithms were critically examined, focusing on issues such as insufficient labeled data, lack of model transparency, computational inefficiency, and poor adaptability to evolving threat patterns. The analysis further explored how these challenges varied across application domains like healthcare, smart homes, and financial systems. In response to these limitations, the reviewed studies proposed a range of strategies, including federated learning, transfer learning, explainable AI techniques, and ensemble methods. The effectiveness of these strategies was evaluated based on empirical results, with specific attention to improvements in model accuracy, interpretability, and resilience to adversarial attacks. This evaluation contributes to a deeper understanding of how ML can be refined and adapted for more effective privacy threat classification in diverse contexts.

4 Background and key concepts

4.1 Privacy Treats

Privacy threats encompass a range of risks that compromise the confidentiality and integrity of personal data across various domains. In the context of identity verification, particularly with facial recognition systems, a comprehensive privacy threat model has been developed using the LINDDUN framework. This model identifies potential threats such as unauthorized data access and inference attacks, which could lead to the exposure of sensitive biometric information[25].

In the realm of federated learning within Internet of Things (IoT) environments, privacy threats like inference attacks, poisoning attacks, and eavesdropping have been identified. Defensive measures such as Differential Privacy and Secure Multi-Party Computation are being explored to mitigate these threats, ensuring data privacy while maintaining the utility of the machine learning models[26].

Additionally, a study on mobile user data privacy threats highlights concerns arising from physical, application-based, network-based, and web-based threats. The research emphasizes the need for comprehensive security measures to address these multifaceted risks in mobile ecosystems [27].

These studies underscore the evolving landscape of privacy threats and the continuous need for innovative models and strategies to safeguard personal data across various technologies and platforms.

4.2 Privacy Treat Classification

Privacy threat classification is the process of systematically identifying, analyzing, and categorizing risks that may compromise the confidentiality and control of personal or sensitive data. According to recent studies published in research [28, 29], this classification plays a critical role in the design of privacy-aware systems by guiding the application of appropriate safeguards based on data sensitivity and threat models. For example, researchers have used techniques such as BERT-based models and local differential privacy mechanisms to classify and protect data according to their level of sensitivity [30]. Additionally, privacy taxonomies have been proposed to categorize threats based on adversarial behavior, data context, and potential impact. These efforts help organizations comply with data protection regulations and improve user trust by addressing specific privacy risks in a targeted manner [31, 32].

4.3 Machine Learning Algorithms

- Support Vector Machine (SVM)
  Support Vector Machine (SVM) has been widely used in privacy threat classification for its capacity to handle high-dimensional data. For instance, studies such as Rama et al. [33] reported excellent performance in detecting intrusions within Internet of

Medical Things (IoMT) networks using SVM, achieving over 99% accuracy. Similarly, in social media, Zhang et al. [34].Patel and Doshi [35]utilized SVM to classify user privacy fatigue with 78% accuracy. SVM is also a popular choice in financial privacy modeling due to its strong decision boundaries. However, SVM's limitations include high computational complexity for large datasets and difficulty in parameter tuning and feature scaling. Furthermore, its lack of interpretability makes it less suitable for sensitive applications that require model transparency [36–38].

- Logistic Regression (LR)
  Logistic Regression (LR) is commonly used in privacy-related classification tasks due to its simplicity and interpretability. It has been applied effectively in healthcare to detect anomalous patient data [39]and in IoT for identifying privacy-related threats [40]. Despite its strengths in handling binary classification and offering insights into feature importance, LR is constrained by its linear decision boundary, making it less effective for complex, non-linear privacy threat patterns. As such, its performance often lags behind more advanced ensemble and deep learning techniques [40, 41].
- Random Forest (RF)
  Random Forest (RF) is widely used in privacy classification due to its robustness, interpretability, and ability to handle both categorical and numerical features. In healthcare, Hussain and et al. [42] demonstrated RF's effectiveness by achieving 99.93% accuracy for intrusion detection in IoMT. It has also shown superior performance in IoT-based federated learning scenarios [43] and finance [44]. RF handles overfitting better than single decision trees and provides feature importance rankings. Nonetheless, it can be computationally expensive, especially with large forests, and becomes harder to interpret as model complexity grows [42–44]
- Decision Trees (DT)
  Decision Trees are frequently employed in privacy applications due to their simplicity and high interpretability. They are used as baseline models in healthcare and IoT, and serve as components in hybrid frameworks like Deep Neural Decision Forests [43, 45]. Their main advantage lies in transparent decision-making processes. However, decision trees are prone to overfitting and often perform poorly on high-dimensional data unless combined with ensemble techniques[46–50].
- Boosting Techniques (AdaBoost, XGBoost, CatBoost)
  Boosting techniques such as AdaBoost, XGBoost, and CatBoost have shown high efficacy in privacy-sensitive domains. For example,Almomani et al. [51] and Hussain et al. [52] reported 99.9% accuracy using XGBoost in IoMT networks. These methods are especially effective in healthcare and IoT environments where high accuracy is critical. Their ability to correct errors from previous iterations makes them highly adaptive. However, boosting methods are computationally expensive and often require significant hyperparameter tuning, which may not be feasible for real-time or low-resource environments [53, 54].
- K-Nearest Neighbors (KNN)
  KNN is often used in privacy threat detection as a benchmark model due to its simplicity and non-parametric nature. In healthcare and IoT datasets, it has demonstrated moderate performance in identifying privacy anomalies [55]. However, KNN's major limitations include scalability issues, sensitivity to noisy data, and

high computational cost in distance calculations for large datasets. Its performance also degrades significantly when dealing with high-dimensional feature spaces [55–57].Ren et al. [58] proposes an efficient and self-recoverable privacy-preserving k-NN classification method, addressing challenges in disease prediction and business forecasting while maintaining data privacy.Alam et al. [59] presents a novel approach for detecting cyber attacks in healthcare systems, highlighting the effectiveness of KNN in identifying privacy anomalies within helthcare networks

- Naïve Bayes (NB)
  Naïve Bayes is another lightweight algorithm applied in healthcare and social media for privacy threat detection. It is effective in classifying user privacy levels and textual data due to its probabilistic foundation [60–62]. The primary advantage is its efficiency and speed. However, NB relies on the assumption of feature independence, which is rarely valid in complex real-world privacy datasets. This often results in lower accuracy compared to other models[63–67].
- Deep Neural Networks (DNNs)
  DNNs are capable of capturing intricate patterns in privacy-related data, making them suitable for complex tasks in healthcare, IoT, and finance. [68–70] used DNNs in federated learning settings, while Almomani et al. [71] combined them with deep autoencoders for intrusion detection. Despite their high accuracy, DNNs are data- and computation-intensive and lack explainability, which restricts their deployment in privacy-sensitive environments Mohammad et al. [72]. Ali et al. [73] presents a new intrusion detection system (IDS) that analyzes network traffic in medical IoT contexts using advanced machine learning and deep learning methods. The IDS can adapt to evolving cyber threats and effectively detect abnormalities by learning from large datasets. Singh et al. [74] discusses a framework for privacy preservation of IoT healthcare data using deep learning. It highlights the role of machine learning (ML) and deep learning (DL) algorithms in extracting hidden patterns, which can be used by organizations to take business decisions while ensuring data privacy.Zhang et al. [70]introduces a Blockchain-orchestrated deep learning approach for secure data transmission in IoT-enabled healthcare systems. It proposes a novel scalable blockchain architecture to ensure data integrity and secure data transmission by leveraging Zero Knowledge Proof (ZKP) protocols.
- Recurrent Neural Networks (RNNs, LSTM, GRU)

  RNNs, LSTM and GRU, are employed for modeling time-dependent privacy threats. They are effective in detecting sequential anomalies in healthcare [75–77] and IoT networks [78]. With accuracy levels exceeding 99% in some IDS datasets, they excel at capturing temporal dependencies. However, these models require significant resources for training and are often seen as black-box models, which hinders their interpretability [78]. Author1 [79] propose a modular framework employing LSTM-RNN for identifying and mitigating low-rate Distributed Denial of Service (DDoS) threats in Software-Defined Networking (SDN) environments.
- Convolutional Neural Networks (CNNs)

CNNs have been repurposed for structured data privacy detection, particularly in hybrid models [80, 81]. Almomani et al. [82] demonstrated high performance using CNN-based deep decision forests. CNNs are also useful in analyzing user-generated content on social media for privacy violations. However, CNNs are typically resource-heavy and lack interpretability, making them challenging to apply in sensitive domains without post-hoc explanation tools. Liu et al. [83] introduces a feedforward-designed CNN (FF-CNN) framework that enhances interpretability and reduces training complexity, making it suitable for privacy-sensitive applications.[84] presents a privacy-preserving CNN framework that enables secure classification and retrieval of large-scale encrypted images, including medical images, using homomorphic encryption.Wang et al. [85] proposes a neural network-based classification hierarchy scheme for privacy data, improving the efficiency and accuracy of privacy threat detection.

5 Results

This section organizes the findings of state-of-the-art studies addressing three research questions.

5.1 What are the exsisting ML algorithms in privacy threat classification?

Based on the analysis of 102 papers on machine learning (ML) algorithms for privacy threat classification, various algorithms have been widely utilized. The most prominent algorithms are Random Forest (RF) and Support Vector Machines (SVM), which together comprise over 40% of the research in this area. RF, known for its robustness and high accuracy in anomaly detection, is frequently used across multiple domains, including healthcare, IoT, and finance. SVM, with its strength in handling linear data classification, is also a popular choice for detecting privacy threats in various environments. Ensemble Methods, including XGBoost, AdaBoost, and CatBoost, account for a significant portion of the research, contributing to enhanced accuracy in complex datasets by combining multiple weak models. Deep Learning models, particularly Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN), are increasingly being applied for their ability to capture complex patterns and dependencies in sequential data, such as IoT traffic or healthcare datasets. Other models like K-Nearest Neighbors (KNN), Naïve Bayes (NB), and Logistic Regression (LR) are often used as baselines, providing simpler solutions with lower computational cost but less accuracy in comparison to more complex models. Together, these algorithms reflect the diversity of approaches being employed to address privacy and security challenges across various domains, each with its strengths and limitations depending on the data and context. These research was conducted using various datasets as 6

The 6 presents a summary of public datasets used for privacy threat classification across various domains. The datasets include a diverse range of privacy threat types, from network intrusion to data leakage and adversarial attacks. For example, CICIDS 2017, a network traffic dataset, focuses on intrusion and data leakage, making

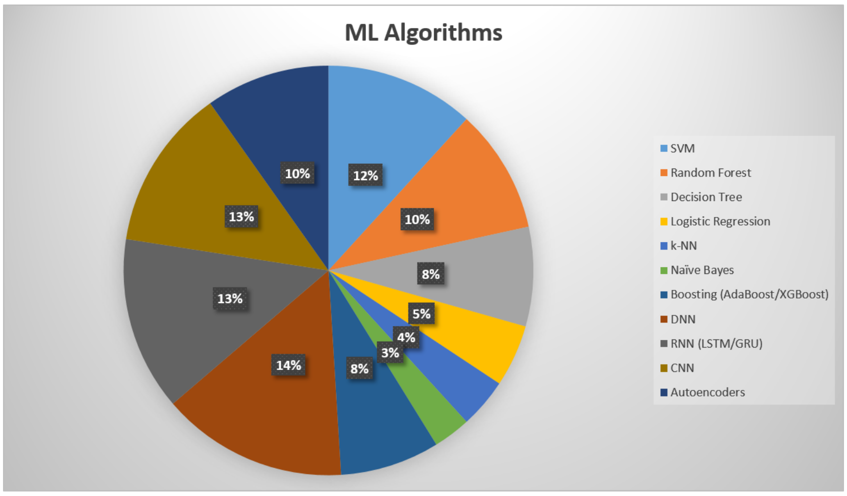

Fig. 2 ML Algorithms

Table 6 Summary of Public Datasets for Privacy Threat Classification

Dataset Name	**Domain**	**Privacy Threat Type**	**Link**	**Year**	**Reference**
CICIDS 2017	Network Traffic	Intrusion, Data Leakage	CICIDS 2017	2017	Sharafaldin et al. [86]
NSL-KDD	Network Traffic	Intrusion, Unauthorized Access	NSL-KDD	2009	Tavallaee et al. [87]
TON_IoT	IoT Devices and Systems	IoT Privacy Attacks, Data Breaches	TON_IoT	2020	Alrashdi et al. [88]
Labeled Faces in the Wild (LFW)	Biometric (Images)	Biometric Privacy Leakage	LFW Dataset	2007	Huang et al. [89]
Twitter Privacy Dataset	Social Media (Text)	Information Disclosure, Profiling	Access via request / papers	2018	Binns et al. [90]
Adversarial Privacy Datasets	Adversarial ML Images	Adversarial Attacks on Privacy	CleverHans	2016	Papernot et al. [91]

it a popular choice for research in detecting privacy-related security breaches in network systemsSharafaldin et al. [86]. Similarly, the NSL-KDD dataset, primarily used for intrusion and unauthorized access detection, has been widely adopted for evaluating intrusion detection systemsTavallaee et al. [87]. In the domain of IoT devices, the TON_IoT dataset is essential for studying IoT privacy attacks and data breachesAlrashdi et al. [88]. The Labeled Faces in the Wild (LFW) dataset, a well-known biometric dataset, focuses on biometric privacy leakage, offering researchers insights into image-based privacy concernsHuang et al. [89]. The Twitter Privacy Dataset, which contains social media text, is used for studying information disclosure and profiling issues, offering a unique perspective on privacy threats in social networksBinns et al. [90]. Finally, the Adversarial Privacy Datasets provided by the CleverHans project are crucial for understanding adversarial attacks in machine learning, particularly in the context of privacy protectionPapernot et al. [91]. These datasets are instrumental in advancing research on privacy threat classification, each contributing to the understanding and mitigation of privacy risks in different contexts.

5.2 What are the limitations of ML algorithms in privacy threat classification?

Machine learning (ML) algorithms have become instrumental in classifying and mitigating privacy threats due to their pattern recognition capabilities. However, significant limitations undermine their deployment, particularly in privacy-sensitive domains. One critical issue is their vulnerability to privacy attacks, such as membership inference, model inversion, and property inference attacks. These techniques allow adversaries to determine if a data point was part of a training set, reconstruct sensitive features, or infer hidden attributes [92, 93]. Such vulnerabilities expose privacy risks even when raw data is not directly accessible.

To mitigate these risks, Privacy-Preserving Machine Learning (PPML) techniques like differential privacy and homomorphic encryption have been introduced. While these methods provide formal privacy guarantees, they often result in decreased model utility and increased computational costs, making them difficult to scale in real-time applications [94, 95]. Furthermore, incorporating PPML methods into existing ML pipelines demands significant system redesign and expert knowledge.

Data-related challenges also limit ML effectiveness in privacy classification. Legal and ethical constraints restrict access to comprehensive, high-quality datasets in domains such as healthcare and finance. Often, available datasets are imbalanced or sparsely labeled, leading to poor generalization. Even anonymized data is prone to re-identification attacks, where attackers can link anonymized data back to individuals [96].

Interpretability is another persistent concern. Most deep learning models, particularly those applied in high-stakes areas, operate as black boxes, offering little insight into their decision-making processes [97]. This undermines user trust and makes it difficult to meet regulatory mandates, such as the General Data Protection Regulation (GDPR), which emphasize transparency and accountability in algorithmic decision-making [98].

Adversarial examples represent an additional risk. These are subtly altered inputs that can cause ML models to make incorrect predictions, revealing the fragility of models to small perturbations [99]. The development of robust defenses against such attacks remains a key area of ongoing research.

Moreover, ML systems are challenged by the dynamic nature of privacy threats. As adversarial techniques evolve, models must be frequently retrained, which is resource-intensive and not always feasible in real-time deployments. This issue is compounded by concept drift, where the statistical properties of the input data change over time, affecting prediction accuracy[97].

Lastly, regulatory and ethical considerations impose constraints on ML systems. Legal requirements around data consent, fairness, and explainability—mandated by policies like GDPR—can be difficult to operationalize in practice. Furthermore, concerns about algorithmic bias, discrimination, and lack of transparency pose ethical challenges, particularly in socially sensitive applications [92].

5.3 What strategies have been proposed to overcome these limitations, and how effective are they?

Numerous strategies have been proposed to address the limitations of machine learning (ML) algorithms in privacy threat classification, with varying degrees of effectiveness depending on the application domain and technical constraints. These strategies span technical, regulatory, and organizational dimensions, and their success often depends on careful trade-offs between privacy, accuracy, interpretability, and scalability [97].

One prominent strategy is differential privacy (DP), which adds mathematical noise to data or model outputs to prevent the re-identification of individuals in a dataset. DP has been particularly effective in scenarios involving centralized data collection, such as census data or aggregate health statistics. However, the noise introduced can degrade model performance, especially in applications requiring high precision, such as healthcare or fraud detection. Moreover, tuning the privacy budget remains a challenge, as stricter privacy guarantees often come at the cost of utility [100].

Federated learning (FL) is another widely adopted approach, especially in settings where data cannot be centralized due to legal or ethical constraints. FL enables ML models to be trained across decentralized devices or servers holding local data samples, without transferring the data itself. This strategy is gaining traction in smart home systems, mobile applications, and healthcare diagnostics. While FL effectively mitigates data sharing risks, it introduces new vulnerabilities such as model inversion and poisoning attacks. Communication costs and heterogeneity of local data distributions also impact model convergence and performance [101].

Advanced training and robust learning techniques have been used to defend against privacy attacks such as model extraction or membership inference. These methods involve training ML models with adversarial examples or using regularization techniques to increase resilience to perturbations. While promising, these strategies often increase computational costs and may reduce model generalizability, making them less suitable for resource-constrained environments like IoT devices [102, 103].

To improve interpretability and compliance, researchers have advocated for the integration of explainable AI frameworks into privacy classification models. Techniques like SHAP, LIME, and saliency maps provide insights into model decisions, fostering trust and aiding in auditing processes. Although useful, these tools do not inherently preserve privacy and can sometimes leak sensitive information through model explanations. Therefore, explainability must be carefully balanced with privacy safeguards [104–106].

At the algorithmic level, privacy-preserving data mining methods such as homomorphic encryption, secure multiparty computation (SMC), and anonymization have been explored. Homomorphic encryption allows computations on encrypted data, ensuring that raw data remains protected throughout processing. SMC enables collaborative computation without revealing individual inputs. However, these techniques are often computationally expensive and challenging to scale to real-time applications or large datasets[107–109].

From a governance perspective, privacy-by-design (PbD) has emerged as a strategic framework encouraging organizations to embed privacy principles at the early stages of ML system development. PbD promotes practices like data minimization, role-based access control, and continuous auditing. While PbD improves privacy assurance and compliance, it requires cultural shifts and investment in privacy engineering skills, which may not be feasible for all organizations[110–112].

Lastly, regulation and policy interventions have played a crucial role in shaping the use of privacy-enhancing ML techniques. Regulations like GDPR, HIPAA, and CCPA have spurred innovation in technical solutions by mandating data protection and transparency. These frameworks provide a legal foundation for enforcing privacy but may lag behind technological advancements, creating compliance ambiguities [113–115].

In conclusion, while a diverse set of strategies exists to mitigate the limitations of ML in privacy threat classification, their effectiveness is context-dependent. Trade-offs between security, usability, and performance are inevitable, and hybrid approaches that combine multiple techniques often yield the best results. A multidisciplinary collaboration between AI researchers, legal experts, and domain practitioners is essential to advance effective and ethical privacy-preserving ML systems [116–118].

6 Discussion

During the review process, a number of challenges associated with machine learning-based privacy threat classification were identified and critically analyzed. One of the prominent issues is the sparsity and incompleteness of labeled threat data, which significantly affects the performance of traditional classification models. Across the literature, a chronological shift is evident in how researchers have attempted to mitigate these limitations, transitioning from conventional machine learning approaches to more sophisticated deep learning and hybrid frameworks.

Before 2018, privacy threat classification was predominantly addressed through rule-based and traditional machine learning methods using handcrafted features extracted from structured datasets. Techniques such as decision trees, support vector machines (SVMs), k-nearest neighbors, and ensemble models were commonly employed [119–121]. These methods relied heavily on static features and lacked adaptability to evolving privacy threat landscapes. Furthermore, issues such as data imbalance, concept drift, and low feature diversity limited the generalizability and robustness of these classifiers. Researchers also explored basic neural networks and fuzzy-based models to improve decision boundaries, but their performance was often constrained by insufficient training data and low-quality feature representations.

Following 2018, the field saw a noticeable transition toward deep learning-driven threat classification, propelled by the need to address the shortcomings of traditional models in handling complex and high-dimensional data. A key advancement during this period has been the adoption of feature enrichment and representation learning methods. For example, profile enrichment through integration of contextual metadata, behavioral features, device and network attributes, and multi-source log data has significantly enhanced input data quality. Approaches such as multi-view learning, attention mechanisms, and representation fusion allow models to capture intricate patterns and dependencies among diverse feature spaces, thereby improving detection accuracy, adaptability, and coverage[33, 122].

In these enriched settings, deep learning models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and autoencoders have demonstrated improved capacity to detect subtle and emerging threats. These models outperform classical algorithms in high-dimensional and dynamic environments, particularly when augmented with attention layers or combined with domain-specific ontologies. Multi-Layer Perceptrons (MLPs), especially when integrated with trust estimation or uncertainty modeling, further improve performance in sparse and imbalanced datasets.

A major improvement in the post-2018 era is the increased emphasis on transparency and interpretability—an essential requirement in privacy-sensitive applications. While early approaches focused mainly on detection accuracy, recent studies have introduced explainable AI (XAI) techniques and uncertainty-aware learning to enhance user trust and regulatory compliance. Multi-criteria threat scoring, review-based data augmentation, and rule-extraction methods have been proposed to clarify classification outcomes and support decision-making processes.

Evaluation of privacy threat classifiers has traditionally relied on metrics such as precision, recall, F1-score, Area Under the Curve (AUC), and Matthews Correlation Coefficient (MCC). However, these metrics often overlook critical aspects like explainability, coverage of unseen threats, and real-time responsiveness—factors now gaining attention due to increased demand for deployable and scalable solutions.

Although deep learning approaches have gained traction in handling data sparsity, class imbalance, and high variability, they still face challenges such as adversarial attacks, high computational overhead, and limited generalizability across domains. Multiview and federated learning have shown promise in mitigating some of these issues by leveraging data from decentralized and heterogeneous sources while preserving privacy. Despite this, their adoption in real-world threat classification systems remains limited, signaling a need for more empirical studies.

In conclusion, while substantial strides have been made in applying machine learning and deep learning to privacy threat classification, significant challenges remain. The integration of robust feature enrichment, explainability, and scalable architectures remains critical for future research. Continued efforts to blend hybrid learning frameworks with interpretability tools will be essential in advancing privacy-aware, trustworthy, and high-performance classification systems.

7 Conclusion

In conclusion, the evolving landscape of privacy threat classification using machine learning reveals a complex interaction between traditional supervised learning methods, feature enrichment strategies, and the growing dominance of deep learning models. While conventional machine learning approaches have laid the foundation for early threat detection, their limitations in handling high-dimensional, sparse, and dynamic threat data have led to the rise of more advanced techniques. The integration of enriched contextual features—combined with the learning capabilities of deep architectures—offers promising avenues for improving accuracy, robustness, and transparency in classification outcomes. However, achieving a balanced understanding of performance metrics such as interpretability, scalability, and coverage remains crucial. As deep learning models become increasingly central to privacy-aware systems, there is a pressing need for further exploration into explainability, cross-domain generalization, and trustworthy AI mechanisms to address emerging privacy threats effectively.

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